Southern New Hampshire University

CS 255 Business Requirements Document

DriverPass | Robert Wilburn | Southern New Hampshire University

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The client for this project is DriverPass, a new driver-training company owned by Liam. Liam identified a gap in the market for preparing students to pass the driving test at their local department of motor vehicles (DMV), after observing that a large number of applicants fail that test.
- The purpose of this project is to design a system that supports the complete DriverPass business model: online classes, online practice tests, in-person instruction on DMV rules and policies, and scheduled on-the-road training with a company driver and vehicle.
- The client wants the system to be web-based and cloud-hosted so that authorized users can reach their data from any computer or mobile device, and so that DriverPass is not responsible for managing backups or server security itself.
- The client wants to be able to download reports and other information so that the data can be worked on offline in a tool such as Excel. Records may only be created or modified while the user is online, which prevents duplicate data and redundancy across servers.
- The system must serve five distinct types of users: the owner (Liam), the IT officer (Ian), the secretary, the drivers and trainers, and the customers. Each of these has a different set of rights within the system.
- The client requires full accountability. Every reservation that is created, modified, or canceled must be traceable to the person who performed the action, and the owner must be able to print an activity report showing that history.
- The deliverable for this phase of the project is a documented design, consisting of this business requirements document along with the use case diagrams, activity diagrams, and class diagram that follow it, to be reviewed with DriverPass before development begins.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- Problem: Liam observed that a large number of people fail the driving test at the DMV. He believes the cause is inadequate preparation, since students often study on their own without structured practice and without a reliable way to know whether the material they are studying still matches current DMV requirements.
- Proposed solution: DriverPass will sell training packages that combine online course content, online practice tests, in-person instruction on DMV rules and policies, and behind-the-wheel training with a company trainer. The system being designed is what makes that offering possible.

- Three packages are offered today, and the system must remain flexible enough to support changes later. Package One is six hours of in-car training with a trainer. Package Two is eight hours plus an in-person lesson covering DMV rules and policies. Package Three is twelve hours, the in-person lesson, and access to the online class with all content, material, and practice tests.
- All driving sessions are two hours long, so a package's total hours are divided across multiple sessions. Package One, for example, consists of three separate two-hour sessions.
- Reservation and scheduling component: customers reserve lessons by day and time, either online through their own account or by calling or visiting the office so that the secretary can enter the reservation for them. DriverPass operates ten cars and each car has an assigned driver, so every reservation must record which student is matched with which driver, which car, and at what time.
- Registration and customer records component: registration captures first name, last name, address, city, state, ZIP code, phone number, and email, along with payment details consisting of credit card number, expiration date, and security code. It also captures a pickup location and a drop-off location, which defaults to the same address as the pickup location.
- Online course and testing component: the system delivers online class material and practice tests to customers whose package includes them, and tracks each test by test name, time taken, score, and status, where status may be not taken, in progress, failed, or passed.
- Security and user management component: role-based rights for each type of employee, with the IT officer holding full administrative access so that he can reset the password of a user who has forgotten it or block access for an employee who has left the company. Customers must also be able to reset their own passwords automatically without staff involvement.
- Auditing and reporting component: the system records which user created, modified, or canceled each record and when, and produces a printable activity report so that the owner can determine who is responsible when a problem occurs.
- DMV compliance component: a connection to the DMV so that new rules, policies, and sample questions can be delivered to DriverPass, with a notification sent whenever an update arrives. This keeps the practice tests aligned with what the DMV actually requires.
- Package administration component: the owner needs the ability to disable a package so that no additional customers can register for it. Adding or removing entire modules will still require a developer or systems analyst and is deferred to a future release.
- Interface component: a cloud-hosted web front end built to the layout the owner sketched, including an online test progress panel, a student information panel, driver notes, a special needs field, driver and student photographs, an input form for student information, a contact page, and a way to contact the student.
- Infrastructure component: cloud hosting with a supporting database, where backup and security are handled by the hosting provider so that DriverPass can concentrate on running the business with minimal technical problems.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- The system shall allow a customer to create an account, log in with validated credentials, and reset a forgotten password automatically without contacting DriverPass staff. Measure: a customer completes a self-service password reset with no staff involvement.

- The system shall allow customers to schedule, modify, and cancel driving lesson reservations online, and shall allow the secretary to perform the same three actions on a customer's behalf by phone or in person. Measure: all three actions succeed from both the customer account and the secretary account.
- The system shall create each reservation as a two-hour session tied to exactly one student, one driver, one car, and one date and time. Measure: no car and no driver can be assigned to two overlapping reservations.
- The system shall support the three current training packages and shall allow the owner to disable any package so that no new customers can register for it. Measure: a disabled package no longer appears as a selectable option during registration.
- The system shall deliver online class content and practice tests to customers enrolled in a package that includes them. Measure: only customers holding Package Three can open the online class.
- The system shall display each customer's test progress showing test name, time taken, score, and a status of not taken, in progress, failed, or passed. Measure: every test a customer has started or completed appears with exactly one of the four statuses.
- The system shall enforce role-based access for the owner, IT officer, secretary, driver, and customer, so that each role can reach only the functions assigned to it. Measure: an account in one role cannot reach a function reserved for another role.
- The system shall give the IT officer full administrative access to reset any account password and to block access for a terminated employee. Measure: a blocked account is denied login on its next attempt.
- The system shall record the user and timestamp for every record that is created, modified, or canceled, and shall produce a printable activity report from that record. Measure: any reservation can be traced to the user who last changed it and the time of that change.
- The system shall receive rule, policy, and sample question updates from the DMV and shall notify an administrator whenever an update is received. Measure: an administrator receives a notification for each DMV update.
- The system shall allow authorized users to download reports in a format that can be opened and worked on in Excel. Measure: a downloaded report opens in Excel with its data intact.
- The system shall operate as a cloud-hosted web application reachable from any standard browser on a computer or mobile device, with backup and security provided by the hosting service. Measure: the same account functions from a desktop browser and from a mobile browser.
- The system shall store a complete student profile including first name, last name, address, city, state, ZIP code, phone, email, pickup location, drop-off location, special needs, and a student photograph, and shall store driver notes containing lesson time, start hour, end hour, and driver comments. Measure: every field from the client's sketch is present on the student page.
- The system shall restrict record creation and modification to online sessions only, so that records cannot be edited offline and later produce duplicate or conflicting data. Measure: no create or update action is available while the user is disconnected.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system shall run as a web-based application hosted in the cloud, reachable through a standard web browser. There is no installed desktop client, which is what allows the owner to reach his data from any computer or mobile device.
- The system shall support the current released versions of the major browsers, including Chrome, Firefox, Safari, and Edge, and shall render correctly on both desktop and mobile screen sizes so that customers can book a lesson from a phone.
- Page loads and standard queries, such as opening a student record or viewing the reservation calendar, shall complete in under three seconds under normal load. Reservation creation and cancellation shall confirm to the user within five seconds.
- The system shall support all DriverPass staff working concurrently along with the customer traffic generated by ten cars' worth of scheduled lessons, and shall be sized so that additional capacity can be added as the customer base grows rather than requiring redesign.
- The system shall be available 99.9 percent of the time, measured monthly, because customers book lessons at all hours through the website rather than only during office hours. Planned maintenance shall occur in a low-traffic window and shall be announced in advance.
- The system shall receive security patches from the hosting provider as they are released, and application updates shall be deployed on a scheduled cycle, with any update that interrupts service performed outside business hours.
- The system shall check for DMV rule, policy, and sample question updates on a recurring schedule, and shall notify an administrator as soon as an update is received so that practice test content does not fall behind current DMV requirements.
- The system shall produce downloadable reports in a format compatible with Excel, and report generation shall complete without locking other users out of the system while it runs.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- Because the system is delivered through a browser, it shall be operating system independent on the client side. Windows, macOS, iOS, Android, and Linux users shall all reach the same application without a platform-specific version being built for each.
- The system shall be hosted on a cloud platform rather than on servers located at the DriverPass office. The IT officer stated that DriverPass does not want to manage backup and security itself, so those responsibilities belong to the hosting provider.
- The back end requires a relational database to store customers, employees, drivers, cars, reservations, packages, test results, driver notes, and the audit log. The relational model fits this data because reservations, drivers, cars, and customers relate to one another in well-defined ways that must be enforced.

- The database shall enforce referential integrity so that a reservation cannot exist without a valid customer, driver, and car, which prevents orphaned bookings that would otherwise appear on a schedule with no one assigned to them.
- The system requires an application server or equivalent cloud service to host the business logic layer, which is where role-based security and scheduling rules are enforced rather than in the browser.
- The system requires an interface to the DMV so that rules, policies, and sample questions can be received. Because the exchange format is controlled by the DMV rather than by DriverPass, the integration shall be built as a separate component that can be modified without altering the rest of the system.
- The system requires a payment processing service to handle credit card data. Card numbers, expiration dates, and security codes shall be handled through that service rather than stored directly by DriverPass, which reduces both risk and compliance burden.
- The system requires an email or messaging service to deliver password reset links, reservation confirmations, and the notification sent when the DMV publishes an update.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- Each user shall be identified by a unique system-generated user ID that is never reused, along with a unique username. Two customers with the same first and last name shall remain distinct records, so the name shall never be used as an identifier.
- Every account shall carry a role of owner, IT officer, secretary, driver, or customer, and that role shall determine the functions the account can reach. The role, not the individual, defines the permissions.
- Usernames and email addresses shall be treated as case insensitive at login, so that a customer who capitalizes their email address is not locked out of their own account. Passwords shall be case sensitive, since case contributes to password strength.
- Data entry fields shall be validated at entry. Phone numbers, ZIP codes, email addresses, and credit card expiration dates shall be checked for correct format, and dates and times for a reservation shall be checked to confirm they fall in the future and within operating hours.
- Reservations shall be validated against existing bookings before they are saved, so that the system rejects an attempt to assign a driver or a car to two lessons at the same time rather than silently creating a conflict.
- Test scores shall be recorded numerically and each test shall carry exactly one status of not taken, in progress, failed, or passed, so that a customer's progress panel cannot display two conflicting states for the same test.
- The system shall notify the IT officer when repeated failed login attempts occur on an account, when a DMV update is received, when a scheduled backup does not complete, and when the connection to the payment service or the DMV interface fails.
- The system shall write an entry to the audit log for every create, modify, and cancel action, recording the user, the action, the record affected, and the timestamp, so that the printable activity report the owner requested is accurate rather than reconstructed after the fact.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- User accounts shall be added, modified, deactivated, and removed entirely through the administrative interface, with no code change and no developer involvement. The IT officer stated that he needs to block access immediately when an employee leaves, which is not possible if account changes require a release.
- Roles and the rights attached to them shall be stored as data rather than written into the application code, so that adjusting what a secretary or a driver may do is a configuration change rather than a rebuild.
- Drivers, cars, and the pairing between them shall be maintained through the interface, so that DriverPass can add an eleventh car or reassign a driver without contacting the development team.
- The owner shall be able to disable a training package so that no additional customers can register for it, while customers already enrolled in that package keep their existing reservations and content access.
- Adding or removing entire modules from the system will continue to require a developer or systems analyst. This was discussed directly with the client and agreed as a future release rather than part of the current build, so the architecture should keep modules loosely coupled to make that later work practical.
- Because the application is delivered through the browser, operating system and device updates on the client side require no action from DriverPass. Browser version changes shall be handled by building to published web standards rather than to the behavior of one browser.
- Platform updates on the server side, including operating system patches, database updates, and hosting infrastructure changes, shall be applied by the cloud provider, which is the arrangement the IT officer asked for.
- The IT officer requires full administrative access across all accounts and system settings, including the ability to reset any password, block any account, assign roles, and view the complete audit log. This is the highest privilege level in the system and shall itself be logged.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- A user shall log in with a unique username and a password before reaching any function of the system. No customer information, reservation, or report shall be viewable without an authenticated session.
- Passwords shall meet a minimum strength standard and shall be stored as salted hashes rather than as recoverable text, so that the stored values are useless to anyone who obtains a copy of the database.
- Staff and administrative accounts shall additionally require multifactor authentication, because those accounts can reach every customer record in the system and a single stolen password should not be enough to expose them.

- All traffic between the client and the server shall be encrypted in transit using TLS, and customer data shall be encrypted at rest. Payment details shall be handled through a payment processing service so that card numbers, expiration dates, and security codes are not stored by DriverPass.
- After a defined number of consecutive failed login attempts, the account shall be locked automatically and the attempt shall be recorded in the audit log with a notification to the IT officer. This is what prevents an automated brute force attempt from continuing indefinitely.
- A locked account shall be released either by the IT officer or by a time-based lockout period, so that a customer who simply forgot their password is not permanently shut out while an attacker is still blocked.
- A customer who forgets their password shall be able to reset it automatically through a time-limited, single-use link sent to the email address on their account, with the new password never displayed or emailed in readable form.
- The IT officer shall be able to reset the password of any account and to block access for an employee who has left the company, with both actions taking effect immediately and both being written to the audit log.
- Sessions shall time out after a period of inactivity, which matters because the secretary works at a shared office workstation where an unattended session would otherwise remain open.
- Every create, modify, and cancel action shall be logged with the user and timestamp, and the log shall be readable but not editable, since an audit trail that can be altered cannot establish who is responsible when something goes wrong.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall validate user credentials at login and deny access when those credentials do not match an active account.
- The system shall allow a customer to create an account and register by entering first name, last name, address, city, state, ZIP code, phone number, and email address.
- The system shall collect payment information at registration, consisting of credit card number, expiration date, and security code, and shall process that information through a payment service.
- The system shall capture a pickup location for each customer and a drop-off location that defaults to the pickup location and may be changed by the customer.
- The system shall allow a customer to select one of the three available training packages during registration.
- The system shall allow the owner to disable a training package so that no additional customers can register for it.
- The system shall allow a customer to reserve a driving lesson by selecting a day and a time through their online account.
- The system shall allow the secretary to create, modify, and cancel a reservation on behalf of a customer who calls or visits the office.
- The system shall allow a customer to modify or cancel an existing reservation through their online account.
- The system shall create every driving lesson reservation as a two-hour session.

- The system shall assign one driver and one car to each reservation and shall record which customer is matched with that driver, car, date, and time.
- The system shall prevent a driver or a car from being assigned to more than one reservation at the same time.
- The system shall track the number of training hours a customer has used and the number remaining under their package.
- The system shall provide online class content and practice tests to customers whose package includes them.
- The system shall record the results of each practice test, including test name, time taken, score, and status.
- The system shall set each test status to not taken, in progress, failed, or passed, and shall display that status on the customer's test progress panel.
- The system shall allow a driver to enter notes for a completed lesson, including lesson time, start hour, end hour, and comments.
- The system shall display driver notes on the student's record so that the customer and staff can review a trainer's feedback.
- The system shall store a special needs entry for each student and display it to the assigned driver.
- The system shall store and display a photograph of the student and a photograph of the assigned driver.
- The system shall allow a customer to reset a forgotten password automatically through a link sent to the email address on their account.
- The system shall allow the IT officer to reset the password of any account in the system.
- The system shall allow the IT officer to block access for an employee who is no longer with the company.
- The system shall assign a role of owner, IT officer, secretary, driver, or customer to every account and shall restrict available functions according to that role.
- The system shall record the user and timestamp for every record that is created, modified, or canceled.
- The system shall generate a printable activity report showing who created, modified, or canceled each reservation and when.
- The system shall allow authorized users to download reports in a format that can be opened in Excel.
- The system shall receive rule, policy, and sample question updates from the DMV.
- The system shall notify an administrator when a DMV update is received.
- The system shall provide a contact page through which a customer can contact DriverPass.
- The system shall provide a means for DriverPass staff to contact a student.
- The system shall prevent records from being created or modified while the user is offline, so that duplicate data is not written to different servers.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The interface shall be a cloud-hosted web application reached through a browser, with a responsive layout so that the same pages work on a desktop monitor, a tablet, and a phone.

Customers book lessons on their own time and often from a mobile device, so mobile support is a requirement rather than an enhancement.

- The interface shall serve five types of users, each seeing only what their role allows: the owner, the IT officer, the secretary, the drivers, and the customers.
- The owner needs a management view showing business activity across the company, access to the printable activity report identifying who created, modified, or canceled each reservation, the ability to disable a training package, and the ability to download reports for offline work in Excel.
- The IT officer needs a full administrative view covering all accounts and system settings, including password resets, blocking access for departed employees, assigning roles and rights, and reviewing the complete audit log.
- The secretary needs a customer service view that supports entering a new registration taken over the phone, searching for an existing customer, and creating, modifying, or canceling reservations on a customer's behalf while speaking with them.
- Drivers need a lesson-focused view showing their assigned schedule, the student for each lesson, the pickup and drop-off locations, the student photograph, any special needs entry, and a form for entering lesson notes with lesson time, start hour, end hour, and comments.
- Customers need a self-service view supporting account registration, package selection, booking and modifying lessons, viewing their reservation history and remaining hours, opening the online class and practice tests, reviewing test progress and driver feedback, resetting a forgotten password, and contacting DriverPass.
- The main student page shall follow the layout the client sketched: a logo across the top, an online test progress panel in the upper left, a student information panel in the upper right holding first name, last name, address, city, state, ZIP code, phone, and email, a driver notes panel in the lower left, and a lower right area containing the special needs entry with the driver photograph and the student photograph beneath it.
- The online test progress panel shall list each test with test name, time taken, score, and status, where status is shown as not taken, in progress, failed, or passed, so that a customer can see at a glance what remains before their DMV appointment.
- The driver notes panel shall present lesson history in a table with columns for lesson time, start hour, end hour, and driver comments.
- The interface shall include an input form for student information that can be completed by the student directly or by the secretary during a phone call, using the same fields in both cases so that staff and customers are not working from different versions of the form.
- The interface shall include a contact page for customers to reach DriverPass, and a means for staff to contact a student about a scheduled lesson.
- The interface shall provide clear feedback on user actions, confirming a successful reservation, highlighting invalid input on a form and prompting the user to correct it, and displaying an understandable message rather than a technical error when an action fails.
- The interface shall follow consistent navigation, terminology, and layout across pages, and shall be designed to meet accessibility standards so that customers using a screen reader or keyboard navigation can complete the same tasks as any other customer.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- It is assumed that DriverPass customers and staff have reliable internet access and a current web browser, since the system is delivered entirely through the web and provides no offline mode for creating or changing records.
- It is assumed that the DMV is willing and able to provide an interface for delivering rule, policy, and sample question updates. The client asked for this connection, but the interview did not confirm that the DMV offers such a service or what format it would use.
- It is assumed that DriverPass will contract with an established payment processing service, so that credit card numbers, expiration dates, and security codes are handled by that provider rather than stored in the DriverPass database.
- It is assumed that the cloud hosting provider will handle backup, disaster recovery, and infrastructure security, since the IT officer stated that DriverPass does not want to manage these itself.
- It is assumed that each of the ten cars has one assigned driver and that a driving session always occupies a full two hours, including travel to the pickup location, since the client described lessons only in two-hour blocks.
- It is assumed that lessons are scheduled only within DriverPass business hours and that the secretary or owner will maintain the calendar of available hours, as the interview did not specify operating hours.
- It is assumed that customers have a valid email address, since automatic password reset and reservation confirmation both depend on email delivery.
- It is assumed that the three training packages described in the interview are the complete current offering and that pricing will be supplied by the client before the interface is built, as no prices were discussed.
- It is assumed that DriverPass operates in a single geographic area served by one DMV, since the client referred to the DMV in the singular and multiple jurisdictions would introduce differing rules per region.
- It is assumed that the online class content and practice test questions will be supplied by DriverPass and the DMV, and that this project covers the system that delivers that content rather than the authoring of the content itself.

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- The system cannot create or modify records while a user is offline. This was raised in the interview as a deliberate decision, because allowing offline changes would produce duplicate data across servers and create redundancy, but it does mean the system is unusable during an internet outage.
- Adding or removing entire modules will require a developer or systems analyst and cannot be performed by a nontechnical user. The owner can disable a package, but restructuring what the packages contain is deferred to a future release.
- The system depends on external services it does not control, including the cloud host, the payment processor, and the DMV interface. An outage or a change in any of those services affects DriverPass directly, and the DMV interface in particular depends on cooperation that has not yet been confirmed.

- Scheduling capacity is limited by physical resources rather than by software. With ten cars and ten drivers, the system can only book as many concurrent lessons as there are available car and driver pairs, so demand beyond that cannot be met regardless of design.
- The project schedule runs from January 22 to May 10 and allows two days for customer approval on March 10 and 11. Every development task after that date depends on that approval, so a request for significant modifications at the approval meeting would push the entire remaining schedule.
- The build phase is sequential. The interface, the database link, the business logic, and testing each depend on the previous task finishing, which means a delay in any one of them moves the delivery date directly.
- Staffing is limited and several tasks run concurrently in late February and early March, with different team members assigned to the activity diagrams, the interface research, and the class diagram. Losing a team member during that window would remove the concurrency the schedule depends on.
- The budget for this project was not discussed in the interview, so cost constraints on hosting, the payment service, and third-party integrations are unknown and could narrow the options available during design.
- Testing is allocated ten days near the end of the schedule. That is sufficient for planned functional testing, but it leaves little room to absorb defects found late without affecting the delivery and sign-off dates.
- The system as designed addresses only the needs described in this interview. The client stated that no future features have been identified yet, so capabilities that emerge after launch will require additional analysis and a separate release.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.

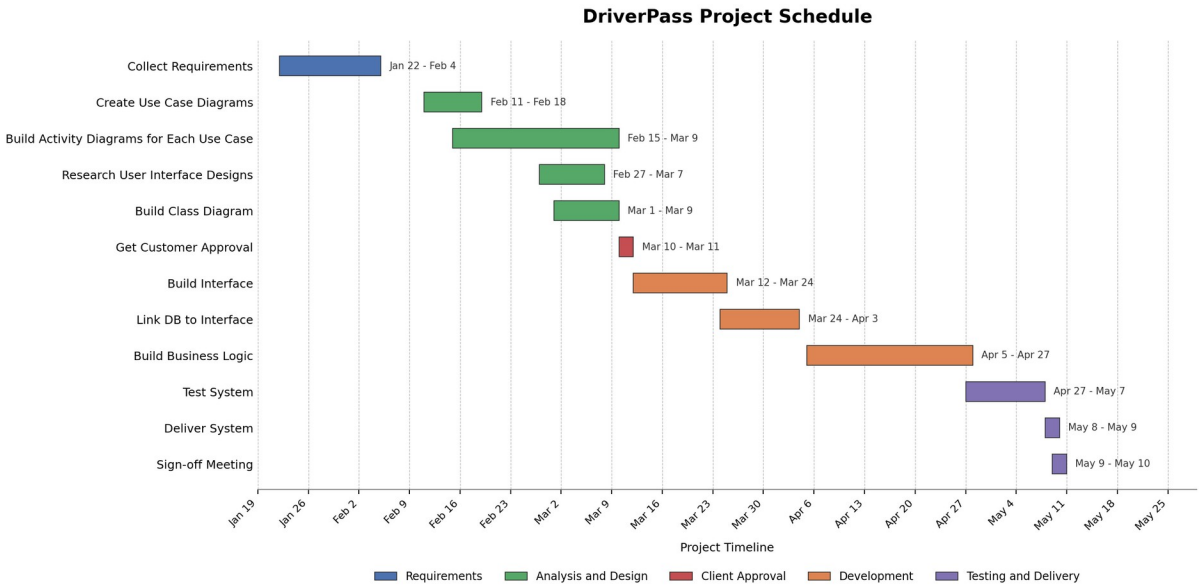


Figure 1. DriverPass project schedule, January 22 through May 10, built from the task list and dependencies agreed in the schedule planning meeting.

- The schedule reflects the task list and dependencies agreed by the project lead and the systems analyst. Requirement collection ran for fourteen days beginning January 22. Use case diagram work begins February 11 rather than immediately afterward because the intervening week is a holiday week.
- Three tasks run concurrently through late February and early March because they have no dependency on one another: the activity diagrams, the user interface research assigned to Toni and Clark, and the class diagram assigned to John, who returns from vacation on March 1. Running these in parallel is what allows the design phase to finish by March 9.
- Customer approval on March 10 and 11 is the gate for the entire build phase. No development task begins until the client has reviewed and approved the design work.
- The build phase is strictly sequential. The interface takes twelve days, linking the database to the interface takes nine, and the business logic layer covering security, roles, and rights takes twenty-two. Each task depends on the one before it, so any delay moves every date that follows.
- System testing runs from April 27 to May 7, followed by delivery on May 8 and 9 and the sign-off meeting on May 9 and 10. Tasks are color coded by phase so that the handoff points between requirements, design, approval, development, and delivery are visible at a glance.